

Note:

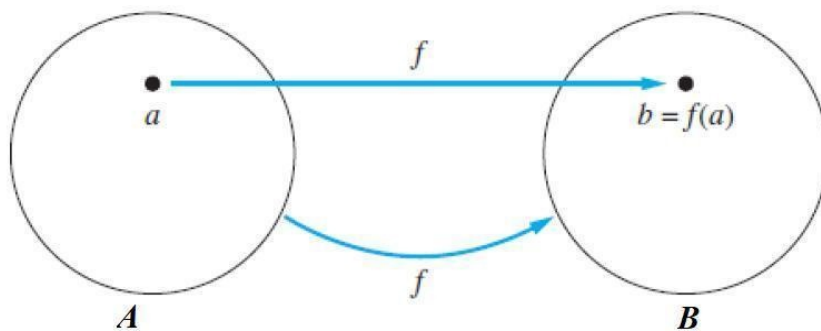
Definition

A function from a set A into a set B , denoted $f : A \rightarrow B$ is a relation from A into B such that each element of A is related to exactly one element of the set B .

- Functions are sometimes also called Maps or Mappings.
- We say that A is the domain of f and B is the codomain of f .
- The range of f is the set of all images of elements of B .

$$R(f) = \{f(a) \in B \mid a \in A\} \subseteq B$$

- For a function, It is customary to use the $y = f(x)$ notation instead of the xfy notation.
- If f maps an element of the domain to zero elements or more than one element of the co-domain, then f is not well-defined.

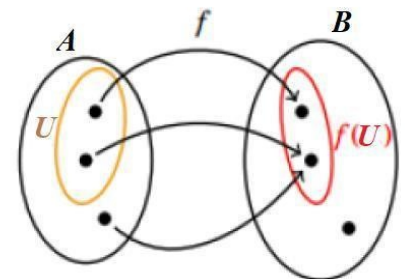


When $f(a) = b$, we say that b is the image of a and a is a preimage of b .
The domain of f is given as

$$\mathcal{D}(f) = \{x \mid \exists y \in A, y = f(x)\}$$

Direct Image and Inverse Image

Direct Image : Let f be a function from A to B and let U be a subset of A . The image of U under the function f is the subset of B that consists of the images of the elements of U . We denote the image of U by $f(U)$, and



$$f(U) = \{f(x) \mid x \in U\}$$

Example : Let $A = \{a, b, c, d\}$, $B = \{\alpha, \beta, \gamma, \epsilon\}$; and define the function $f : A \rightarrow B$, with $f(a) = f(b) = \beta$, $f(d) = \gamma$, $f(c) = \epsilon$, then

$$f(\{b\}) = \{\beta\}, f(\{a, c\}) = \{\beta, \epsilon\}, f(\{a, c, d\}) = \{\beta, \epsilon, \gamma\}.$$

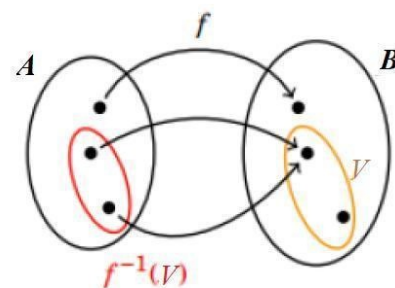
Note: $y \in f(U) \iff \exists x \in U, y = f(x)$
($y = f(x)$ for some $x \in U$)

Inverse Image : Let f be a function from A to B and let V be a subset of B . The inverse image of V under the function f is the subset of A that consists of the preimages of the elements of V . We denote the inverse image of V by $f^{-1}(V)$, and

$$f^{-1}(V) = \{x \in A \mid f(x) \in V\}$$

- $f^{-1}(V)$ is the part of A formed by the preimages of the elements of V . We can therefore write

$$x \in f^{-1}(V) \iff f(x) \in V$$



Example Let $f : \mathbb{R} \longrightarrow \mathbb{R}$ such that $f(x) = 4x + 5$ and $V = \{x \in \mathbb{R} \mid x > 0\}$

$$\begin{aligned} f^{-1}(V) &= \{x \in A \mid f(x) \in V\} \\ &= \{x \in \mathbb{R} \mid f(x) > 0\} \\ &= \{x \in \mathbb{R} \mid 4x + 5 > 0\} \\ &= \{x \in \mathbb{R} \mid x > \frac{-4}{5}\} \end{aligned}$$

Exercises

- Let $f(x) := 1/x^2$, $x \neq 0$, $x \in \mathbb{R}$.
 - Determine the direct image $f(E)$ where $E := \{x \in \mathbb{R} : 1 \leq x \leq 2\}$.
 - Determine the inverse image $f^{-1}(G)$ where $G := \{x \in \mathbb{R} : 1 \leq x \leq 4\}$.
- Let $f(x) := x^2$ for $x \in \mathbb{R}$, and let $E := \{x \in \mathbb{R} : -1 \leq x \leq 0\}$ and $F := \{x \in \mathbb{R} : 0 \leq x \leq 1\}$. Show that $E \cap F = \{0\}$ and $f(E \cap F) = \{0\}$, while $f(E) = f(F) = \{y \in \mathbb{R} : 0 \leq y \leq 1\}$. Hence $f(E \cap F)$ is a proper subset of $f(E) \cap f(F)$. What happens if 0 is deleted from the sets E and F ?
- Let f and E, F be as in Exercise 02. Find the sets $E \setminus F$ and $f(E) \setminus f(F)$ and show that it is *not* true that $f(E \setminus F) \subseteq f(E) \setminus f(F)$.
- Show that if $f : A \rightarrow B$ and E, F are subsets of A , then $f(E \cup F) = f(E) \cup f(F)$ and $f(E \cap F) \subseteq f(E) \cap f(F)$.
- Show that if $f : A \rightarrow B$ and G, H are subsets of B , then $f^{-1}(G \cup H) = f^{-1}(G) \cup f^{-1}(H)$ and $f^{-1}(G \cap H) = f^{-1}(G) \cap f^{-1}(H)$.

- * { 06. (a) Show that if $f : A \rightarrow B$ is injective and $E \subseteq A$, then $f^{-1}(f(E)) = E$.
 (b) Show that if $f : A \rightarrow B$ is surjective and $H \subseteq B$, then $f(f^{-1}(H)) = H$.

Definition. Two functions $f : A \rightarrow B$ and $g : C \rightarrow D$ are **equal** if and only if

1. $A = C$,
2. $B = D$,
3. $f = g$. ($f = g$ if and only if $f(x) = g(x)$ for every $x \in A$.)

Note:

Equality of Functions:

Two functions f and g defined on the same domain D are said to be **equal** if and only if $f(x) = g(x)$ for all $x \in D$.

Example:

Show that the functions $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = |x|$ for all $x \in \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ defined by $g(x) = \sqrt{x^2}$ for all $x \in \mathbb{R}$ are equal.

Operations on Functions

Let f be a function with domain A and g be a function with domain B . Let $c \in \mathbb{R}$. Then,

(i) the **sum** $f + g$ is defined by

$$\begin{aligned}(f + g)(x) &= f(x) + g(x) \text{ for all } x \in D, \text{ where} \\ D = D(f + g) &= \{x | x \in D(f) \text{ and } x \in D(g)\} \\ &= \{x | x \in A \text{ and } x \in B\} \\ &= A \cap B.\end{aligned}$$

(ii) the **scalar product** cf is defined by

$$\begin{aligned}(cf)(x) &= c \cdot f(x) \text{ for all } x \in D, \text{ where} \\ D = D(cf) &= A.\end{aligned}$$

(iii) the **difference** $f - g$ is defined by

$$\begin{aligned}(f - g)(x) &= (f + (-g))(x) \text{ for all } x \in D, \text{ where} \\ D = D(f - g) &= D(f + (-g)) \\ &= \{x | x \in D(f) \text{ and } x \in D(-g)\} \\ &= A \cap B.\end{aligned}$$

(iv) the **product** fg is defined by
 $(fg)(x) = f(x)g(x)$ for all $x \in D$, where
 $D = D(fg) = \{x | x \in D(f) \text{ and } x \in D(g)\}$
 $= \{x | x \in A \text{ and } x \in B\}$
 $= A \cap B.$

(v) the **quotient** $\frac{f}{g}$ is defined by

$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$ for all $x \in D$, where

$D = D\left(\frac{f}{g}\right) = \{x | x \in A \cap B \text{ and } g(x) \neq 0\}$

(vi) Let $f: A \rightarrow C$ and $g: B \rightarrow D$ be two functions such that $A \subseteq B$ and g be a functions. Then the

composition $f \circ g: B \rightarrow C$ is a function defined by

$(f \circ g)(x) = f(g(x))$ for all $x \in D(f \circ g)$, where

$D(f \circ g) = \{x | x \in B \text{ and } g(x) \in A\}.$

Example:

For the functions $f(x) = 2x^2 - 4x + 1$ and $g(x) = 7x$, find $f + g, f - g, fg,$

$\frac{f}{g}$, and their domains.

Solution:

The domain of $f(x) = 2x^2 - 4x + 1$ is $(-\infty, \infty)$. The domain of $g(x) = 7x$ is $(-\infty, \infty)$. The intersection of the domains of f and g is $(-\infty, \infty)$.

$$(f + g)(x) = f(x) + g(x) = 2x^2 - 4x + 1 + 7x = 2x^2 + 3x + 1$$

The domain of $f + g$ is $(-\infty, \infty)$.

$$(f - g)(x) = f(x) - g(x) = 2x^2 - 4x + 1 - 7x = 2x^2 - 11x + 1$$

The domain of $f - g$ is $(-\infty, \infty)$.

$$(fg)(x) = f(x)g(x) = (2x^2 - 4x + 1)(7x) = 14x^3 - 28x^2 + 7x$$

The domain of fg is $(-\infty, \infty)$.

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} = \frac{2x^2 - 4x + 1}{7x}$$

The domain of $\frac{f}{g}$ is $(-\infty, 0) \cup (0, \infty)$.

Example:

For the functions $f(x) = 3x + 1$ and $g(x) = 4x + 5$, evaluate $(f + g)(1),$

$(f - g)(2), (fg)(-1), \left(\frac{f}{g}\right)(-2),$ and $(f \circ g)(3).$

Theorem. Suppose $f : A \rightarrow B$ and $g : B \rightarrow C$. Then $g \circ f : A \rightarrow C$, and for any $a \in A$, the value of $g \circ f$ at a is given by the formula $(g \circ f)(a) = g(f(a))$.

Scratch work

Before proving this theorem, it might be helpful to discuss the scratch work for the proof. According to the definition of function, to show that $g \circ f : A \rightarrow C$ we must prove that $\forall a \in A \exists! c \in C((a, c) \in g \circ f)$, so we will start out by letting a be an arbitrary element of A and then try to prove that $\exists! c \in C((a, c) \in g \circ f)$.

To prove existence, we should try to find a $c \in C$ such that $(a, c) \in g \circ f$. For uniqueness, we should assume that $(a, c_1) \in g \circ f$ and $(a, c_2) \in g \circ f$, and then try to prove that $c_1 = c_2$.

Proof. Let a be an arbitrary element of A . We must show that there is a unique $c \in C$ such that $(a, c) \in g \circ f$.

Existence: Let $b = f(a) \in B$. Let $c = g(b) \in C$. Then $(a, b) \in f$ and $(b, c) \in g$, so by the definition of composition of relations, $(a, c) \in g \circ f$. Thus, $\exists c \in C((a, c) \in g \circ f)$.

Uniqueness: Suppose $(a, c_1) \in g \circ f$ and $(a, c_2) \in g \circ f$. Then by the definition of composition, we can choose $b_1 \in B$ such that $(a, b_1) \in f$ and $(b_1, c_1) \in g$, and we can also choose $b_2 \in B$ such that $(a, b_2) \in f$ and $(b_2, c_2) \in g$. Since f is a function, there can be only one $b \in B$ such that $(a, b) \in f$. Thus, since (a, b_1) and (a, b_2) are both elements of f , it follows that $b_1 = b_2$. But now applying the same reasoning to g , since $(b_1, c_1) \in g$ and $(b_1, c_2) = (b_2, c_2) \in g$, it follows that $c_1 = c_2$, as required.

This completes the proof that $g \circ f$ is a function from A to C . Finally, to derive the formula for $(g \circ f)(a)$, note that we showed in the *existence* half of the proof that for any $a \in A$, if we let $b = f(a)$ and $c = g(b)$, then $(a, c) \in g \circ f$. Thus,

$$(g \circ f)(a) = c = g(b) = g(f(a)). \quad \square$$

Definition . Let $f : A \rightarrow B$ and $g : C \rightarrow D$ be functions such that $\text{ran } f \subseteq \text{dom } g$. Then the function $g \circ f : A \rightarrow D$ defined by $(g \circ f)(x) = g(f(x))$ is called the *composition* of f and g .

Note:

$$D(g \circ f) = \{x \mid x \in D(f) \text{ and } f(x) \in D(g)\}.$$

Example

For each of the following cases find, if possible, the domain, codomain and rule of assignment for $g \circ f$ and $f \circ g$:

1. $f : \mathbb{N} \rightarrow \mathbb{R}, f(n) = n + 1; g : \mathbb{R} \rightarrow \mathbb{R}, g(x) = \sqrt[3]{x}$.
2. $f : \mathbb{Z} \rightarrow \mathbb{Z}, f(x) = -x; g : \mathbb{R} \rightarrow \mathbb{R}, g(x) = \sin x$.
3. $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = 3x; g : \mathbb{R} \rightarrow \mathbb{R}, g(x) = \frac{x}{3}$.

4 Let f and g be functions from $\mathbb{R}$ to $\mathbb{R}$ defined by the following formulas:

$$f(x) = \frac{1}{x^2 + 2}, \quad g(x) = 2x - 1.$$

Find formulas for $(f \circ g)(x)$ and $(g \circ f)(x)$.

Lemma1. If f and g are functions such that $\text{ran } f \subseteq \text{dom } g$, then $\text{dom}(g \circ f) = \text{dom } f$, and $\text{codom}(g \circ f) = \text{codom } g$.

Lemma 2. Let $f : A \rightarrow B, g : B \rightarrow C, h : C \rightarrow D$ be functions. Then, $h \circ (g \circ f) = (h \circ g) \circ f$, i.e. the composition of functions is associative.

Proof. Let $p = h \circ (g \circ f)$, and $q = (h \circ g) \circ f$; to show that two functions are equal.

Note:

$$D(g \circ f) = \{x \mid x \in D(f) \text{ and } f(x) \in D(g)\}.$$

$$D(f \circ g) = \{x \mid x \in D(g) \text{ and } g(x) \in D(f)\}.$$

EXAMPLE 8 Let $f: [0, \infty) \rightarrow \mathbb{R}$ and $g: (-\infty, 2] \rightarrow \mathbb{R}$ defined by
 $f(x) = \sqrt{x}$ for all $x \in [0, \infty)$ and $g(x) = \sqrt{2-x}$ for all $x \in (-\infty, 2]$.
 find each function and its domain.

- (a) $f \circ g$ (b) $g \circ f$ (c) $f \circ f$ (d) $g \circ g$

SOLUTION

(a)

$$(f \circ g)(x) = f(g(x)) = f(\sqrt{2-x}) = \sqrt{\sqrt{2-x}} = \sqrt[4]{2-x}$$

The domain of $f \circ g$ is $\{x \mid 2-x \geq 0\} = \{x \mid x \leq 2\} = (-\infty, 2]$.

$$(b) (g \circ f)(x) = g(f(x)) = g(\sqrt{x}) = \sqrt{2-\sqrt{x}}$$

For $\sqrt{x}$ to be defined we must have $x \geq 0$.

For $\sqrt{2-\sqrt{x}}$ to be defined we must have

$2-\sqrt{x} \geq 0$, that is, $\sqrt{x} \leq 2$, or $x \leq 4$.

Thus, we have $0 \leq x \leq 4$, so the domain of

$g \circ f$ is the closed interval $[0, 4]$.

$$(c) (f \circ f)(x) = f(f(x)) = f(\sqrt{x}) = \sqrt{\sqrt{x}} = \sqrt[4]{x}$$

The domain of $f \circ f$ is $[0, \infty)$.

$$(d) (g \circ g)(x) = g(g(x)) = g(\sqrt{2-x}) = \sqrt{2-\sqrt{2-x}}$$

This expression is defined when $2-x \geq 0$,

that is, $x \leq 2$, and $2-\sqrt{2-x} \geq 0$. This

equivalent to $\sqrt{2-x} \leq 2$, or $2-x \leq 4$,

that is, $x \geq -2$. Thus,

domain of $g \circ g$ is the closed interval $[-2, 2]$.

End of D3